

CHEM 5200 Practice Problems

Spin- $\frac{1}{2}$ Formalism and Ladder Operators

How to use these problems. Each part is tagged by how it is meant to be done: **[hand]** = pencil and paper, **[code]** = Python/NumPy, **[explain]** = a few sentences of physical reasoning. Each problem ends with an optional *grad extension* that asks for a proof or a harder step.

Units and tools. Set $\hbar = 1$ throughout (and $m = \omega = 1$ for oscillators), as in the notebooks. Reuse the functions you built in Computational Set 1 (`compute_bra_ket`, `compute_expectation`) and the helpers in the Second Quantization notebook (`annihilation`, `creation`, `position`, `momentum`, `comm`, `expval`).

Notation. The $\hat{S}_z$, $\hat{S}_x$ and $\hat{S}_y$ eigenstates are

$$|\alpha\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\beta\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad |X^\pm\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad |Y^\pm\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Part I Spin- $\frac{1}{2}$

SP1. Measuring spin along a tilted axis

Along a unit vector $\hat{n} = (n_x, n_y, n_z)$, the spin operator is

$$\hat{S}_n = n_x \hat{S}_x + n_y \hat{S}_y + n_z \hat{S}_z.$$

- [hand]** Take $\hat{n} = (3/5, 0, 4/5)$. Check that $\hat{n}$ is a unit vector, then write $\hat{S}_n$ as a 2×2 matrix.
- [code]** Use `la.eigh` on $\hat{S}_n$. Confirm that the eigenvalues are $\pm\hbar/2$, the same as for $\hat{S}_z$. Show that the $+$ eigenvector is proportional to $(3, 1)^T$.
- [hand]** A beam of $|\alpha\rangle$ atoms enters a Stern–Gerlach magnet aligned along this $\hat{n}$. What fraction leaves through the $+$ port?
- [code]** Repeat (b) and (c) for $\hat{n} = (0, 3/5, 4/5)$. Show that the $+$ eigenvector is now $\frac{1}{\sqrt{10}}(3, i)^T$ and that the probability is still 9/10. **[explain]** Why does tilting toward y instead of x leave the probability unchanged?
- [code]** Tilt $\hat{n}$ in the xz -plane by an angle θ , so $\hat{n} = (\sin\theta, 0, \cos\theta)$. Loop θ from 0 to π , compute $P(+)$ for an $|\alpha\rangle$ beam at each angle, and plot it against $\cos^2(\theta/2)$.
- [hand]** $\rightarrow$ **[code]** Keep only the $+$ beam leaving each magnet in a chain.
 - By hand: chain $z \rightarrow x \rightarrow -z$. What fraction survives?
 - With code: use N magnets, each tilted a further π/N from the previous one, starting from z and ending at $-z$. Compute the transmitted fraction for $N = 2, 3, 5, 10, 100$.

- (g) **[explain]** Adding more measurements lets almost every atom end up flipped. Explain why, using the measurement postulate.

Grad extension (optional for undergraduates)

For a general $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, show that $|n^+\rangle = \left(\cos \frac{\theta}{2}, e^{i\phi} \sin \frac{\theta}{2} \right)^T$.

SP2. Expectation values and the Bloch vector

Use $|\psi\rangle = \frac{1}{\sqrt{5}}|\alpha\rangle + \frac{2i}{\sqrt{5}}|\beta\rangle$ and $\hbar = 1$.

- (a) **[code]** Use `compute_expectation` to find $\langle\sigma_x\rangle$, $\langle\sigma_y\rangle$ and $\langle\sigma_z\rangle$.
- (b) **[hand]** Collect the three numbers into the **Bloch vector** $\vec{r} = (\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle)$. (You will use $\vec{r}$ again in SP4.) Compute its length $|\vec{r}|$. **[explain]** What does it mean for the arrow to point mostly along $+y$ and partly along $-z$?
- (c) **[code]** The uncertainty of an observable is $\Delta\sigma_x = \sqrt{\langle\sigma_x^2\rangle - \langle\sigma_x\rangle^2}$. *Hint:* first compute σ_x^2 as a matrix. What do you notice? Compute $\Delta\sigma_x\Delta\sigma_y$ and compare it with the uncertainty bound $|\langle\sigma_z\rangle|$.
- (d) **[code]** Generate 1000 random normalized spinors. For each one compute $|\vec{r}|$, $\Delta\sigma_x\Delta\sigma_y$ and $|\langle\sigma_z\rangle|$.
- Confirm that $|\vec{r}| = 1$ every time and that the product never falls below the bound.
 - Check numerically that $(\Delta\sigma_x\Delta\sigma_y)^2 = \langle\sigma_z\rangle^2 + \langle\sigma_x\rangle^2\langle\sigma_y\rangle^2$.
- (e) **[explain]** Use the formula in (d) to say when the bound is met exactly. Why did the state in (a) meet it?

Grad extension (optional for undergraduates)

Prove the identity in (d) for any pure state. (*Hint:* use $\sigma_i^2 = \mathbb{I}$ and $|\vec{r}| = 1$.)

SP3. Two conventions for “raising”

This problem starts from the *Note on conventions* in the spin- $\frac{1}{2}$ lecture and in Computational Set 4. Set $\hbar = 1$.

- (a) **[code]** Build the lecture’s $\hat{S}_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\hat{S}_- = \hat{S}_+^\dagger$. Apply each to $|\alpha\rangle$ and $|\beta\rangle$ (a numerical check of Question 13).
- (b) **[code]** Build the cavity notebook’s $\sigma_+ = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $\sigma_- = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Apply each to $|g\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|e\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Use `np.allclose` to confirm that $\sigma_+ = \hat{S}_+^\dagger = \hat{S}_-$ and $\sigma_- = \hat{S}_+$.
- (c) **[code]** Compute $[\hat{S}_+, \hat{S}_-]$ and $[\sigma_+, \sigma_-]$ and compare them with $\hat{S}_z$ and σ_z .

- (d) **[explain]** The two sets are the same matrices, yet the commutators differ in sign. Explain in one or two sentences why both are correct.
- (e) **[hand]** Take $\hat{H} = -\frac{\omega}{2}\sigma_z$. Find the energies of $|g\rangle$ and $|e\rangle$ and say which is lower. What is the σ_z eigenvalue of the ground state, and why is the minus sign needed?
- (f) **[code]** (Toward ladder operators.)
- Compute $\sigma_+\sigma_-$ and compare it with $\hat{a}^\dagger\hat{a}$ for a cavity truncated to 2 states. What does each count?
 - Compute $\sigma_+\sigma_+$, then $\hat{a}^\dagger\hat{a}^\dagger$ for a cavity truncated to 3 states. One is zero and one is not. In one sentence, explain the difference between a spin, whose ladder really ends, and a truncated oscillator.

Grad extension (optional for undergraduates)

The lecture's $\sigma_+ = \frac{1}{2}(\sigma_x + i\sigma_y)$ gives $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Which combination of σ_x and σ_y gives the cavity notebook's σ_+ ?

SP4. Polarized and unpolarized beams

Set $\hbar = 1$. For a state $|\psi\rangle$ the **density matrix** is $\rho = |\psi\rangle\langle\psi|$. A beam that is a mixture of states $|\psi_k\rangle$ with probabilities p_k has $\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|$. The probability of leaving through the $+$ port of a magnet along $\hat{n}$ is $\langle n^+ | \rho | n^+ \rangle$.

- (a) **[code]** Build $\rho_X = |X^+\rangle\langle X^+|$. Confirm that $\langle n^+ | \rho_X | n^+ \rangle$ equals $|\langle n^+ | X^+ \rangle|^2$ for magnets along z , along x , and along the $\hat{n}$ from SP1(a).
- (b) **[code]** Atoms leaving the oven are unpolarized: $\rho_{\text{oven}} = \frac{1}{2}\mathbb{I}$. Compute $P(+)$ for magnets along z , x , y and SP1's $\hat{n}$. **[explain]** Which single magnet tells the oven beam and the $|X^+\rangle$ beam apart?
- (c) **[code]** In SP2 the Bloch vector was $\vec{r} = (\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle)$. For a density matrix, compute its components as $r_x = \text{Tr}(\rho\sigma_x)$, and likewise for y and z (for a pure state this gives the same numbers as SP2).
- Compute $\vec{r}$ for ρ_X and for ρ_{oven} .
 - Verify numerically that $\rho = \frac{1}{2}(\mathbb{I} + r_x\sigma_x + r_y\sigma_y + r_z\sigma_z)$ for both.
- (d) **[code]** $\rightarrow$ **[explain]** Compute $\text{Tr}(\rho^2)$ for both beams. What does the length of $\vec{r}$ tell you about a beam?
- (e) **[code]** Build two beams: a 50/50 mixture of $|\alpha\rangle$ and $|\beta\rangle$, and a 50/50 mixture of $|X^+\rangle$ and $|X^-\rangle$. Show that their density matrices are identical. **[explain]** Why can no experiment tell these two beams apart?
- (f) **[code]** Take $\hat{H} = \omega_0\hat{S}_z$, for which $U(t) = \text{diag}(e^{-i\omega_0 t/2}, e^{i\omega_0 t/2})$.
- Evolve $|X^+\rangle$ and plot the components of $\vec{r}(t)$.
 - Plot the probability $P(X^+, t)$ of passing an x -magnet at time t .

- Show that $U\rho_{\text{oven}}U^\dagger = \rho_{\text{oven}}$. **[explain]** Why does the unpolarized beam not evolve?

Grad extension (optional for undergraduates)

Integrate the Liouville–von Neumann equation $\dot{\rho} = -i[\hat{H}, \rho]$ with a fourth-order Runge–Kutta step and compare with the exact result in (f). This is the same integrator you will use in Computational Set 4.

Part II Ladder Operators

Recall $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$, $\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$, $\hat{x} = \frac{1}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger)$, $\hat{p} = \frac{i}{\sqrt{2}}(\hat{a}^\dagger - \hat{a})$, $\hat{H}_0 = \hat{a}^\dagger\hat{a} + \frac{1}{2}$ (with $\hbar = m = \omega = 1$).

LP1. IR selection rules from ladder operators

- (a) **[hand]** Find $\hat{a}|2\rangle$, $\hat{a}^\dagger|2\rangle$, $\hat{a}^\dagger\hat{a}|2\rangle$ and $\hat{a}\hat{a}^\dagger|2\rangle$. Show that $\hat{a}\hat{a}^\dagger|2\rangle - \hat{a}^\dagger\hat{a}|2\rangle = |2\rangle$; this is $[\hat{a}, \hat{a}^\dagger] = 1$ at work.
- (b) **[hand]** Find $\hat{a}^\dagger\hat{a}^\dagger|1\rangle$.
- (c) **[hand]** Write $\hat{x}|n\rangle$ in terms of number states. For which m can $\langle m|\hat{x}|n\rangle$ be nonzero? State this as the IR selection rule.
- (d) **[code]** Build `position(6)` and print it. Where do the nonzero entries sit? Tabulate $|\langle n+1|\hat{x}|n\rangle|^2$ for $n = 0$ to 3.
- (e) **[explain]** Per molecule, the $1 \rightarrow 2$ “hot band” absorbs twice as strongly as the $0 \rightarrow 1$ fundamental. For HCl ($\tilde{\nu} = 2886 \text{ cm}^{-1}$, $kT \approx 208.5 \text{ cm}^{-1}$ at 300 K), estimate the fraction of molecules in $n = 1$. Why are hot bands still rarely seen at room temperature?
- (f) **[code]** Real dipoles are not exactly linear in x . Take $\mu(x) = x + 0.1x^2$. Compute $\langle 2|\mu|0\rangle$ and $\langle 1|\mu|0\rangle$, and the overtone-to-fundamental intensity ratio $|\langle 2|\mu|0\rangle|^2/|\langle 1|\mu|0\rangle|^2$. Which term in μ makes the overtone allowed?

Grad extension (optional for undergraduates)

Use $[\hat{a}, \hat{a}^\dagger] = 1$ to show that $[\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger$. Conclude that $\hat{a}^\dagger|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue $n + 1$.

LP2. Anharmonicity by counting ladder paths

- (a) **[hand]** Expand $\langle 0|(\hat{a} + \hat{a}^\dagger)^4|0\rangle$ into ordered strings of four operators. Keep only the strings that leave $|0\rangle$ and return to it without dropping below the bottom rung (write $U = \hat{a}^\dagger$ for a step up and $D = \hat{a}$ for a step down). Weight each surviving string by its $\sqrt{n}$ factors and add them. What is $\langle 0|\hat{x}^4|0\rangle$?
- (b) **[explain]** Using the same picture, explain why $\langle n|\hat{x}^3|n\rangle = 0$ for every n .
- (c) **[code]** Compute $\langle n|\hat{x}^4|n\rangle$ for $n = 0$ to 3, using a basis large enough to obey the padding rule of §1.6. Compare with $(6n^2 + 6n + 3)/4$.
- (d) **[code]** Take $\hat{H} = \hat{H}_0 + \lambda\hat{x}^4$. The first-order energy correction is $\lambda\langle 0|\hat{x}^4|0\rangle$, so $E_0 \approx \frac{1}{2} + \frac{3}{4}\lambda$. Compare this estimate with the lowest eigenvalue from `eigvalsh` in a large basis, for $\lambda = 0.01$ and $\lambda = 0.1$. When does first order stop being good enough?

- (e) **[code]** At $\lambda = 0.1$, compute E_0 for basis sizes $N = 2, 3, 4, 5, 6, 8, 12$, building $\hat{x}^4$ from the truncated $\hat{x}$ matrix. **[explain]** The values jump above and below the converged answer. Using the padding rule, explain why.
- (f) **[explain]** Why does a cubic term $\hat{x}^3$ do nothing to the energy at first order, even though it is the leading anharmonic term in a real bond?

Grad extension (optional for undergraduates)

Derive $\langle n | \hat{x}^4 | n \rangle = \frac{1}{4}(6n^2 + 6n + 3)$ by counting and weighting all return paths from rung n .

LP3. Coherent states: where the classical field comes from

The amplitude is called z so it is not confused with the spin state $|\alpha\rangle$.

- (a) **[code]** Build $|z\rangle = e^{-|z|^2/2} \sum_n \frac{z^n}{\sqrt{n!}} |n\rangle$ with $z = 2$ in an $N = 40$ basis. Check that it is normalized and that $\hat{a} |z\rangle = z |z\rangle$.
- (b) **[code]** Compute $\langle \hat{N} \rangle$ and plot $P(n) = |\langle n | z \rangle|^2$.
- (c) **[code]** Compute $\langle \hat{x} \rangle$, Δx and Δp . Compare with §1.9.
- (d) **[code]** Evolve with $|z(t)\rangle = \sum_n c_n e^{-i(n+1/2)t} |n\rangle$. Plot $\langle \hat{x}(t) \rangle$ and $\langle \hat{p}(t) \rangle$ and check that they trace the classical orbit. Does Δx change in time?
- (e) **[explain]** In §2.4 every number state gave $\langle n | \hat{E} | n \rangle = 0$. Here $\hat{E} = i(\hat{b}^\dagger - \hat{b}) = \sqrt{2} \hat{p}$, so $\langle \hat{E}(t) \rangle = -4 \sin t$. Why is laser light described by coherent states rather than number states?

Grad extension (optional for undergraduates)

Show directly from the series that $\hat{a} |z\rangle = z |z\rangle$.

LP4. A displaced oscillator and the Franck–Condon progression

When a molecule is electronically excited, the minimum of its bond potential shifts. Model the excited-state vibrational Hamiltonian as $\hat{H}_e = \hat{H}_0 + g(\hat{a} + \hat{a}^\dagger)$.

- (a) **[hand]** Write $g(\hat{a} + \hat{a}^\dagger)$ in terms of $\hat{x}$. Explain why this term shifts the potential minimum sideways, and in which direction for $g > 0$.
- (b) **[code]** Diagonalize $\hat{H}_e$ in an $N = 40$ basis for $g = 1$.
- Show that the levels are still evenly spaced but lowered: $E_n = n + \frac{1}{2} - g^2$.
 - Show that the new ground state $|\psi\rangle$ satisfies $\hat{a} |\psi\rangle = -g |\psi\rangle$, so it is a coherent state (LP3) with $z = -g$, and that $\langle \hat{x} \rangle = -\sqrt{2}g$.

- (c) **[code]** Absorption starts from the ground state of $\hat{H}_0$, which is $|0\rangle$. Compute the Franck-Condon factors $|\langle \psi_n^{(e)} | 0 \rangle|^2$ for $n = 0$ to 4 and compare with the Poisson distribution $e^{-S} S^n / n!$, where $S = g^2$ is the Huang-Rhys factor.
- (d) **[code]** Repeat for $g = \sqrt{2}$ ($S = 2$). Plot both progressions as stick spectra. **[explain]** How does the size of the shift control which vibronic peak is tallest?
- (e) **[explain]** What does the energy shift $-g^2$ mean physically? Connect it to the MgH vibronic-spectra notebook.

Grad extension (optional for undergraduates)

Show that substituting $\hat{a} = \hat{b} - g$ turns $\hat{H}_e$ into $\hat{b}^\dagger \hat{b} + \frac{1}{2} - g^2$.